Statistics & Probability: Complete Beginner Notes

Built from your lecture screenshots, in a learning order. Anything marked ➕ **Added** was missing from the slides and has been filled in so the notes are complete.

Learning Roadmap

1. Basics: what statistics and probability are
2. Averages: mean, median, mode, weighted mean
3. Spread: variance and standard deviation
4. Probability basics: sample space, events, P(E)
5. Types of events
6. Probability rules: complement, addition, multiplication
7. Conditional probability
8. Law of total probability
9. Bayes' theorem
10. Random variables
11. Probability distributions
12. Cheat sheet and practice questions

1. The Big Picture ➕ Added

- **Statistics** works backwards: you have data, and you describe it or draw conclusions from it.
- **Probability** works forwards: you know the rules of a situation, and you predict how likely outcomes are.
- **Population**: the whole group you care about (all students in a country).
- **Sample**: a smaller part of the population you actually measure (200 students).
- **Parameter**: a number describing a population (μ, σ).
- **Statistic**: a number describing a sample ($\bar{x}$, s).

Symbol	Meaning
μ (mu)	population mean
$\bar{x}$ (x-bar)	sample mean

Symbol	Meaning
σ^2 (sigma squared)	population variance
s^2	sample variance
σ (sigma)	population standard deviation
n	number of values
Σ (sigma, capital)	"add up everything"

2. Measures of Central Tendency ("the typical value")

Mean: the average value

Mean = (sum of all values) ÷ (number of values)

$\mu = \Sigma x_i / n$

Median: the middle value

1. Sort the data from smallest to largest.
2. If n is odd, take the middle value.
3. If n is even, average the two middle values.

Mode: the most frequent value

The value that appears most often. A dataset can have no mode, one mode, or several.

Weighted Mean

Used when some values matter more than others (for example, exam marks with different weights).

Weighted mean = $\Sigma(w_i \cdot x_i) / \Sigma w_i$ ➕ Added

Example: Assignment = 80 (weight 30%), Final exam = 90 (weight 70%). Weighted mean = (0.3 × 80 + 0.7 × 90) / (0.3 + 0.7) = (24 + 63) / 1 = **87**

Worked example (we will reuse this data)

Data: 2, 4, 4, 4, 5, 5, 7, 9

- Mean = (2+4+4+4+5+5+7+9) / 8 = 40 / 8 = **5**
- Median = sorted already, the two middle values are 4 and 5, so (4+5)/2 = **4.5**

- Mode = **4** (appears 3 times)

> **Tip:** Outliers (extreme values) pull the mean but barely move the median. For skewed data like salaries, the median is often more honest.

3. Measures of Spread ("how scattered is the data?")

Using **variance** and **standard deviation** we measure how spread out the numbers are from the mean.

Variance (σ^2)

$\sigma^2 = \Sigma(x_i - \mu)^2 / n$

Step by step:

1. Find the mean μ.
2. Subtract the mean from each value: $(x_i - \mu)$.
3. Square each result (so negatives do not cancel positives).
4. Add them all up.
5. Divide by n.

Standard Deviation (σ)

$\sigma = \sqrt{\text{variance}} = \sqrt{[\Sigma(x_i - \mu)^2 / n]}$

It is the square root of variance, so it is in the **same units** as the original data (variance is in squared units).

Worked example with 2, 4, 4, 4, 5, 5, 7, 9 ($\mu = 5$)

x	$x - \mu$	$(x - \mu)^2$
2	−3	9
4	−1	1
4	−1	1
4	−1	1
5	0	0
5	0	0

x	x − μ	(x − μ)²
7	2	4
9	4	16
Sum		**32**

- Variance $\sigma^2 = 32 / 8 = \mathbf{4}$
- Standard deviation $\sigma = \sqrt{4} = \mathbf{2}$

➕ Added: Sample variance (when your data is a sample)

$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$

We divide by **n − 1** instead of n because a sample tends to underestimate the true spread. Here: 32 / 7 ≈ 4.57, so s ≈ 2.14.

➕ Added: Range

Range = maximum − minimum = 9 − 2 = 7 (the simplest measure of spread).

Remember: small σ means values are close to the mean. Large σ means values are widely scattered.

4. Probability: The Basics

Probability is a measure of the **likelihood** that a particular event will occur. It ranges from **0 to 1**.

- 0 means impossible.
- 1 means certain.
- 0.5 means a 50-50 chance.
- You can also write it as a percentage (0.25 = 25%).

Two kinds of probability

1. Theoretical probability: calculated from reasoning, assuming all outcomes are equally likely. Example: a fair coin has P(heads) = 1/2 without flipping it even once.

2. Empirical (experimental) probability: calculated from actual experiments or observations.

P(E) = (number of times E occurred) / (total number of trials) ➕ Added

Example: you flip a coin 100 times and get 56 heads, so empirical P(heads) = 56/100 = 0.56. With more trials, empirical probability moves closer to theoretical (the Law of Large Numbers).

Important terms

- **Sample Space (S)**: all possible outcomes. Rolling a die: S = {1, 2, 3, 4, 5, 6}.
- **Event (E)**: a subset of outcomes. "Rolling an even number": E = {2, 4, 6}.
- **P(E)**: the probability of event E.

The core formula

P(E) = (number of favorable outcomes) / (total number of outcomes) = n(E) / n(S)

Example: P(even number on a die) = 3/6 = **1/2**

5. Types of Events

Certain and impossible

- **Certain event**: always happens. P(S) = 1.
- **Impossible event**: cannot happen. P(∅) = 0.

Intersection and union

- **P(A ∩ B)**: probability that A **and** B both occur (intersection, "overlap").
- **P(A ∪ B)**: probability that A **or** B occurs, meaning at least one (union).

Relationships between two events

Type	Meaning	Example
Mutually exclusive (disjoint)	Cannot happen at the same time	Rolling a 2 and rolling a 5 on one die
Independent	One does not affect the probability of the other	Two separate coin flips
Dependent	One affects the probability of the other	Drawing 2 cards without putting the first back
Complementary	One occurs exactly when the other does not	Passing / failing

> **Common trap:** Mutually exclusive is NOT the same as independent. If two events are mutually exclusive and both have non-zero probability, they are actually **dependent** (if A happens, B definitely cannot).

6. The Three Probability Rules

Rule 1: Complementary Rule

If A is an event, then its complement A^c (A not happening) has probability:

$P(A^c) = 1 - P(A)$

Example: P(rain) = 0.3, so P(no rain) = 1 − 0.3 = **0.7**. Use this when "at least one" is hard: P(at least one head in 3 flips) = 1 − P(no heads) = 1 − $(1/2)^3$ = **7/8**.

Rule 2: Addition Rule (the word "OR")

Mutually exclusive events (cannot happen together):

$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$

Example: P(rolling a 1 or a 2) = 1/6 + 1/6 = **1/3**.

Non-mutually exclusive events (can happen together):

$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$

Why subtract? The overlap gets counted twice when you add, so remove it once.

Example: drawing a card that is a **heart or a king**. P = 13/52 + 4/52 − 1/52 = 16/52 = **4/13** (the king of hearts is in both groups).

Rule 3: Multiplication Rule (the word "AND")

Independent events:

$P(A \text{ and } B) = P(A) \times P(B)$

Example: two heads in two flips = 1/2 × 1/2 = **1/4**.

Dependent events:

$P(A \text{ and } B) = P(A) \times P(B \mid A)$

Example: two aces in a row **without replacement**: 4/52 × 3/51 = 12/2652 = **1/221**.

> **Memory trick:** OR means add (and subtract overlap). AND means multiply.

7. Conditional Probability

The probability of event B occurring **given that** event A has already happened.

$P(B \mid A) = P(A \cap B) / P(A)$

Read "|" as "given". The condition shrinks your sample space to only the cases where A happened.

Example: In a class, 60% study maths (A), and 24% study both maths and physics (A ∩ B). What is the chance a student studies physics, given they study maths? $P(B \mid A) = 0.24 / 0.60 =$ **0.4**

Connection to independence ➕ Added: A and B are independent exactly when $P(B \mid A) = P(B)$.

8. Law of Total Probability

Used to find P(B) by splitting it into several mutually exclusive cases $A_1, A_2, \ldots, A_n$.

$P(B) = P(B \mid A_1) \cdot P(A_1) + P(B \mid A_2) \cdot P(A_2) + \ldots + P(B \mid A_n) \cdot P(A_n)$

Conditions:

- $A_1, A_2, \ldots, A_n$ are **mutually exclusive** and **exhaustive** (exactly one of them must happen, so their probabilities add to 1).
- $P(B \mid A_i)$ is the probability of B given A_i occurred.

Example: A shop gets bulbs from 3 factories.

Factory	Share of supply $P(A_i)$	Defect rate $P(B \mid A_i)$
A_1	50%	2%
A_2	30%	3%
A_3	20%	5%

$P(\text{defective}) = (0.02 \times 0.5) + (0.03 \times 0.3) + (0.05 \times 0.2) = 0.010 + 0.009 + 0.010 =$ **0.029 (2.9%)**

9. Bayes' Theorem

Bayes' theorem lets you **reverse** a conditional probability.

$P(A \mid B) = [\, P(B \mid A) \times P(A) \,] / P(B)$

Where:

- P(A | B): probability of A given B occurred (the answer you want, called the **posterior**)
- P(B | A): probability of B given A occurred (the **likelihood**)
- P(A): probability of A before knowing B (the **prior**)
- P(B): total probability of B (often found with the Law of Total Probability)

➕ **Added: expanded form** (for two cases A and A^c):

$$P(A \mid B) = P(B \mid A)\cdot P(A) \,/\, [\, P(B \mid A)\cdot P(A) + P(B \mid A^c)\cdot P(A^c) \,]$$

Classic example: medical test

- 1% of people have a disease: P(D) = 0.01
- The test is positive for 99% of sick people: P(+ | D) = 0.99
- The test is falsely positive for 5% of healthy people: P(+ | no D) = 0.05

You test positive. What is the chance you actually have the disease?

1. P(+) = (0.99 × 0.01) + (0.05 × 0.99) = 0.0099 + 0.0495 = 0.0594
2. P(D | +) = 0.0099 / 0.0594 ≈ **0.167 (about 16.7%)**

Surprising but true: because the disease is rare, most positive results are false alarms.

10. Random Variables

A **random variable (RV)** is a numerical quantity that depends on the outcome of a random experiment. It is usually written with a capital letter such as X.

- **Discrete RV**: countable values (integers). Number of heads in 10 flips, number of students absent.
- **Continuous RV**: any value in a range (fractions allowed). Height, temperature, time taken.

➕ Added: Expected value (the long-run average of an RV)

Discrete: $E(X) = \Sigma\, x_i \cdot P(x_i)$

Example, one fair die: E(X) = (1+2+3+4+5+6)/6 = **3.5**

➕ Added: Variance of an RV

$$Var(X) = E(X^2) - [E(X)]^2 = \Sigma\, (x_i - \mu)^2 \cdot P(x_i)$$

Example, one fair die: $E(X^2) = 91/6 \approx 15.17$, so $Var(X) = 15.17 - 12.25 \approx$ **2.92**, and $\sigma \approx 1.71$.

11. Probability Distributions

A probability distribution describes how probabilities are assigned to all possible outcomes. It tells you **how likely different outcomes are**.

	Discrete RV	Continuous RV
Function	PMF (probability mass function)	PDF (probability density function)
Gives	P(X = x) directly	Area under the curve = probability
Rule	All probabilities add to 1	Total area under curve = 1
Single point	Can have positive probability	P(X = exact value) = 0

➕ **CDF (cumulative distribution function):** $F(x) = P(X \leqslant x)$, the probability of being at or below x. Works for both types.

Types of probability distributions

Distributions split into two families, matching the two kinds of random variable.

- **Discrete probability distribution**: for a discrete RV, meaning outcomes we can count. Example: **Binomial distribution** (also Bernoulli and Poisson).
- **Continuous probability distribution**: for a continuous RV, meaning the outcome can be any number in a range. Examples: **Uniform distribution** and **Normal distribution**.

Quick test: can you list the outcomes one by one (0, 1, 2, 3...)? Then it is discrete. Do you measure it with a ruler, scale or clock, so decimals are possible? Then it is continuous.

The rest of this section follows that order: discrete first (Bernoulli, Binomial, Poisson), then continuous (Uniform, Normal).

➕ Discrete distributions: Bernoulli, Binomial and Poisson

Bernoulli: one trial, two outcomes (success with probability p, failure with 1 − p).

Binomial: number of successes in n independent trials, each with success chance p.

$P(X = k) = C(n, k) \cdot p^k \cdot (1 - p)^{n-k}$, where $C(n, k) = n! / [k!(n - k)!]$

- Mean = $n \cdot p$
- Variance = $n \cdot p \cdot (1 - p)$
- Example: exactly 3 heads in 5 flips = $C(5,3) \times (1/2)^3 \times (1/2)^2 = 10 \times 1/32 =$ **0.3125**

Poisson: number of events in a fixed interval when events happen at an average rate λ (calls per hour, typos per page).

$P(X = k) = e^{-}\lambda \cdot \lambda^{k} / k!$

- Mean = variance = λ

Continuous distribution 1: Uniform distribution

A uniform distribution is a probability distribution where **all outcomes are equally likely**.

- **Discrete uniform**: coin toss, rolling a die. With n outcomes, P(X = x) = 1/n, so each face of a die has probability 1/6.
- **Continuous uniform**: pick a random number from 0 to 10. Every value in the range is equally likely.

➕ **Formulas for a continuous uniform on the range [a, b]:**

- Height of the curve: f(x) = 1 / (b − a)
- Probability of landing between c and d: P(c ⩽ X ⩽ d) = (d − c) / (b − a)
- Mean = (a + b) / 2
- Variance = $(b - a)^2 / 12$

Example: a random number from 0 to 10 (a = 0, b = 10). P(2 ⩽ X ⩽ 5) = 3/10 = **0.3**. Mean = **5**. Variance = 100/12 ≈ **8.33**.

Because the number is continuous, P(X = exactly 5) = 0. Only ranges have probability.

Continuous distribution 2: Normal distribution

The normal distribution is a continuous probability distribution that is **bell-shaped and symmetric**. We write $X \sim N(\mu, \sigma^2)$ ➕.

- Most values cluster around the mean (μ).
- Values far from the mean are less likely.
- The shape is symmetric around the mean: the left half mirrors the right half.

Characteristics

1. **Mean = Median = Mode**, all sitting at the centre of the curve.
2. **Total area under the curve = 1** (the probabilities of all outcomes add to 100%).
3. **μ = mean** (where the centre is) and **σ = standard deviation** (how spread out the curve is). A small σ gives a tall, narrow bell. A large σ gives a short, wide bell.
4. **Shortcut (68-95-99.7 rule):** about 68% of the data lies within ±1σ of the mean, 95% within ±2σ, and 99.7% within ±3σ.

Range	Share of data	Example with μ = 70, σ = 10
μ ± 1σ	about 68%	60 to 80
μ ± 2σ	about 95%	50 to 90
μ ± 3σ	about 99.7%	40 to 100

On the board, the points are marked μ − 3σ, μ − 2σ, μ − 1σ, μ, μ + 1σ, μ + 2σ, μ + 3σ along the bottom of the bell. Only about 0.3% of values fall beyond 3σ (0.15% in each tail), and the tails never quite touch the axis.

➕ **The formula behind the curve (you rarely compute it by hand):**

```
f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}
```

➕ **Z-score:** how many standard deviations a value x is from the mean. z = (x − μ) / σ. Converting to z lets you use one standard table, the **standard normal** with μ = 0 and σ = 1.

Example: exam mean 70, σ = 10. A score of 90 has z = (90 − 70) / 10 = **2**, so it beats about 97.5% of students. Using the shortcut instead: P(60 ⩽ X ⩽ 80) ≈ 0.68, and P(X > 80) ≈ (1 − 0.68) / 2 = **0.16**.

12. Cheat Sheet (all formulas in one place)

Topic	Formula
Mean	μ = Σx / n
Weighted mean	Σ(w·x) / Σw
Variance (population)	σ² = Σ(x − μ)² / n
Variance (sample)	s² = Σ(x − x̄)² / (n − 1)
Standard deviation	σ = √variance
Basic probability	P(E) = n(E) / n(S)
Empirical probability	P(E) = times E occurred / total trials
Limits	0 ⩽ P(E) ⩽ 1, P(S) = 1, P(∅) = 0
Complement	$P(A^c) = 1 - P(A)$
Addition (mutually exclusive)	P(A ∪ B) = P(A) + P(B)
Addition (general)	P(A ∪ B) = P(A) + P(B) − P(A ∩ B)

Topic	Formula
Multiplication (independent)	$P(A \cap B) = P(A) \cdot P(B)$
Multiplication (dependent)	$P(A \cap B) = P(A) \cdot P(B \mid A)$
Conditional	$P(B \mid A) = P(A \cap B) / P(A)$
Total probability	$P(B) = \Sigma P(B \mid A_i) \cdot P(A_i)$
Bayes	$P(A \mid B) = P(B \mid A) \cdot P(A) / P(B)$
Expected value	$E(X) = \Sigma x \cdot P(x)$
Binomial	$C(n,k)\, p^k\, (1-p)^{n-k}$
Poisson	$e^{-\lambda}\, \lambda^k / k!$
Z-score	$z = (x - \mu) / \sigma$

Which rule do I use? Quick decision guide

- Question says "**not**" or "**at least one**" → complement rule.
- Question says "**or**" → addition rule (subtract overlap if events can both happen).
- Question says "**and**" / "both" / "in a row" → multiplication rule (ask: does the first event change the second?).
- Question says "**given that**" → conditional probability.
- Event can come from several different sources or cases → law of total probability.
- You know $P(B \mid A)$ but want $P(A \mid B)$ → Bayes.

13. Practice Questions

1. Find the mean, median and mode of: 3, 7, 7, 2, 9, 7, 5.
2. Find the variance and standard deviation of: 1, 3, 5, 7.
3. A bag has 4 red and 6 blue balls. What is the probability of drawing a red ball?
4. $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Find $P(A \cup B)$.
5. Two cards are drawn without replacement from a standard deck. What is the probability both are kings?
6. $P(A) = 0.6$, $P(A \cap B) = 0.3$. Find $P(B \mid A)$.
7. A coin is flipped 4 times. What is the probability of exactly 2 heads?

Answers

1. Mean = 40/7 ≈ 5.71; sorted 2,3,5,7,7,7,9 so median = 7; mode = 7.

2. Mean = 4; squared deviations 9+1+1+9 = 20; variance = 20/4 = 5; $\sigma \approx 2.24$.
3. 4/10 = 0.4.
4. 0.5 + 0.4 − 0.2 = 0.7.
5. $4/52 \times 3/51 = 1/221$.
6. 0.3 / 0.6 = 0.5.
7. $C(4,2) \times (1/2)^4 = 6/16 = 0.375$.

Study tip: Do not memorise formulas in isolation. For each one, make up a small example (a die, a coin, a deck of cards) and solve it by hand. After a few examples the formulas stick naturally.